Notas de Combinatória 2

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April 13, 2026

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0 Informações importantes

Devo guardar notas de aula e exercícios interessantes do curso. Por enquanto nenhuma notação muito esotérica apareceu, se sim devo listá-las aqui. O objetivo dessas notas é serem breves, de forma que é possível recuperar formalmente as provas dos teoremas enunciados. Não serei estritamente rigoroso, o que pode levar a alguns erros. A linguagem pode variar intermintentemente entre português e inglês.

O código fonte desse arquivo pode ser encontrado em `https://github.com/hnrq104/comb2/tree/main`.

1 Extremal Set Theory

Lemma 1.1. (LYMB) Let $\mathcal{A} \subset \mathcal{P}([n])$ be an anti-chain, then

$$\sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1} \leqslant 1$$

Proof. (Sketch) Take a random permutation and consider the indicator random variables X_A that the first $|A|$ elements form A. We find that $\mathbb{E} \sum X_A = \sum_{A \in \mathcal{A}} \binom{n}{|A|}^{-1}$, but $\sum X_A \leqslant 1$. $\square$

Corollary 1.2. (Sperner) If $\mathcal{A}$ is an anti-chain, $|\mathcal{A}| \leqslant \binom{n}{n/2}$.

Corollary 1.3. (Littlewood-Offord) Given $a_1, a_2, \dots a_n \in \mathbb{R} \setminus \{0\}$. Consider $x_1, \dots x_n \in \{-1, 1\}$ u.i.i.d r.v's, then $\mathbb{P}(\sum a_i x_i = 0) \leqslant 2^{-n}\binom{n}{n/2}$.

Proof. By simmetry of the x_i, we may assume that all a_i are positive. Now consider the collection $\mathcal{A} := \{A \subset [n] : \sum_{i \in A} a_i - \sum_{i \notin A} a_i = 0\}$. $\mathcal{A}$ is an anti-chain. □

Theorem 1.4. (Erdos-Ko-Rado, Katona) Let $\mathcal{A} \subset \binom{[n]}{k}$ be an intersecting family with $2k < n$, then $|\mathcal{A}| \leqslant \binom{n-1}{k-1}$.

Proof. (Sketch) Take a random circular permutation of $[n]$, let X_A be the indicator of A appearing as an interval on the circle. We find that $\sum X_A \leqslant k$ - partition the intervals that intersect A and use pigeonhole. Calculating expectation and rearranging gives the result. □

Theorem 1.5. (Exercise ERK) There is $n_0(k, l)$ such that if $n \geqslant n_0$ and $\mathcal{A} \subset \binom{[n]}{k}$ is an l-intersecting family, then $|\mathcal{A}| \leqslant \binom{n-l}{k-l}$. Hints:

1. Show that there exists $L \subset [n]$, $|L| = l$ such that $L = A \cap B$ for $A, B \in \mathcal{A}$.
2. Show that either $\mathcal{A} \subset \partial L := \{A \in \binom{[n]}{k} : L \subset A\}$ or $|\mathcal{A}| = O(n^{k-l-1})$.

Proof. For the first step, assume $\mathcal{A}$ is not empty and for every $A, B \in \mathcal{A}$, $|A \cap B| \geqslant l + 1$. Choose any element C from $\mathcal{A}$ and notice that every set in the family must intersect at least with $l + 1$ elements from C, so the number of sets in $\mathcal{A}$ is bounded by

$$|\mathcal{A}| \leqslant \binom{k}{l+1}\binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

So we may assume that there are A, B, L as in the hint. Now, either $\mathcal{A} \subset \partial L$ and we are done, or there exists $C \in \mathcal{A}$ such that $L \not\subset C$. How many such C can exists? Because $A \cap B = L$, $L \not\subset C$, $|A \cap C| \geqslant l$ and $|B \cap C| \geqslant l$ we find that

$$|C \cap (A \cup B)| = |C \cap (A \cap B)| + |C \cap (A \setminus B)| + |C \cap (B \setminus A)| \geqslant l + 1$$

where we used the fact that $l \leqslant |C \cap A| = |C \cap L| + |C \cap (A \setminus B)|$ and so $|C \cap (A \setminus B)| \geqslant 1$. So the number of such C is at most $\binom{|A \cup B|}{l+1}\binom{n-|A \cup B|}{k-l-1} = O(n^{k-l-1})$. And now, fix any such C, how many elements of $\mathcal{A}$ may contain L? Because they all intersect in C an element outside of L, we have at most

$$|C \setminus L|\binom{n-l-1}{k-l-1} = O(n^{k-l-1}).$$

such elements, which completes the proof. □

I really liked Rob's motivation for continuing to study these extremal problems.

(a) An explicit construction that gives $R(k) \geqslant k^C$ for any fixed C!

(b) The chromatic number of $\mathbb{R}^n$. We define $\chi(\mathbb{R}^n)$ as the minimal number of colors k such that there exists a function $f : \mathbb{R}^n \to [k]$ satisfying that for every $x, y \in \mathbb{R}^n$ with $|x - y| = 1$, we have that $f(x) \neq f(y)$. For example, we know by taking a simplex on $\mathbb{R}^n$ that $\chi(\mathbb{R}^n) \geqslant n+1$. It is possible to show (by considering a tesselation by cubes with diameter less than 1) that the number is well defined for any dimension. We will use extremal stuff to prove some order of growth in n of this parameter.

(c) (Borsuk's Conjecture) Show that for every set $X \subset \mathbb{R}^n$ it is possible to decompose it in $n+1$ sets $X_1, \dots X_{n+1}$ s.t. $\operatorname{diam}(X_i) < X$.

Now lets go back to some theorems. The next few ones are nice applications of linear algebra.

Definition 1.6. Given $L \subset [n]$, we say a family $\mathcal{A} \subset \mathcal{P}(n)$ is L-intersecting if for every $A \neq B \in \mathcal{A}$ we have that $|A \cap B| \in L$.

Theorem 1.7. Let $\mathcal{A} \subset \mathcal{P}(n)$ and let $L \subseteq \mathbb{Z}_p$ with $|L| = s$. If $\mathcal{A}$ is L-intersecting and for every $A \in \mathcal{A}$, $A \pmod p \notin L$, then

$$|\mathcal{A}| \leqslant \sum_{i=0}^{s} \binom{n}{i}$$

Proof. Consider the L.I set

$$f_A(X) = \prod_{l \in L} (X \cdot \mathbb{1}_A - l)$$

in $\mathbb{F}_p[x_1, x_2 \dots x_n]$ and calculate the dimension. □

Theorem 1.8. Let $\mathcal{A} \subset \binom{n}{k}$ be L intersecting, then $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$.

Proof. Consider $f_A = \prod_{l \in L}(X \cdot \mathbb{1}_A - l) \in \mathbb{R}[x_1, \dots, x_n]$, this always has degree less than $|L|$ and all monomials (after correcting the exponents) have degree at most $|L|$. So, by the dimension bound (of vectors being L.I), we know that $|\mathcal{A}| \leqslant \binom{n}{\leqslant |L|}$. □

Now let's sketch some applications.

Theorem 1.9. There is a explicit coloring that shows $R(m) \geqslant m^C$ for any constant C. (In our setup) There exists a sequence of graphs on N vertices with no clique or independent set of size $N^{o(1)}$.

Proof. The idea is to consider a graph, such that cliques and independent sets correspond to L intersecting families. So let's fix our $V(K_N) = \binom{[n]}{k}$. And consider a coloring of $E[K_N]$ by making

$$c(A, B) = \begin{cases} \text{R} & \text{if} \quad |A \cap B| \in L \\ \text{G} & \text{if} \quad |A \cap B| \notin L \end{cases}$$

where $L \subset [k-1]$. Now, what's the maximum size of a red clique? We know by [1.8], that it is at most by $\binom{n}{\leqslant |L|}$. So if L is relatively small, we can expect to win something. What about green cliques? Then we turn to the modular theorem [1.7]. If we suppose $k = p^2 - 1$ and let

$L = \mathbb{Z}_p \setminus \{-1\}$. We would have that each green clique satisfies the hypothesis of 1.7, and so has size at most $\binom{n}{\leqslant p-1}$.

So thats the idea, we take $L = \{p-1, 2p-1, \ldots p(p-1)-1\}$ so our coloring has no red clique of size $\binom{n}{\leqslant p-1}$, and no green clique of size $\binom{n}{\leqslant p-1}$.

Suffices to show that if $N = \binom{n}{k}$, then $\binom{n}{\leqslant \sqrt{k}} = N^{o(1)}$. □

Theorem 1.10. $\chi(\mathbb{R}^n) \geqslant \exp(cn)$

Proof. The proof follows similarly, we somehow want to find a finite embeddable graph in $\mathbb{R}^n$ with small independence number. Then $\chi(\mathbb{R}^n) \geqslant \chi(G) \geqslant |V(G)|/\alpha(G)$. So, let's look into our setting again of $V(G) = \binom{[n]}{k}$, and to embedd stuff in $\mathbb{R}^n$ we will consider the indicator functions k-sets in $\mathbb{R}^n$ (which are vectors of n dimensions).

Now we want to forbid a distance between these vectors (which will make our sets independent). So let's forbid a specific distance (which we can always scale to be the right thing) and say

$$E(G) = \{(A, B) : |A \cap B| \neq t\}$$

For some $t \in \mathbb{Z}$. Here as we don't really care about the clique number (only the independence), so we need to bound the size of a independent set. So in order to get the strongest bound, we use theorem [1.7] and suppose $k = 2p - 1$ for some p and $L = t \pmod p$. We get $\alpha(G) \leqslant \binom{n}{1}$. So by taking k very big (say $n/2$) we get the result.

□

Theorem 1.11. (Disproof of Borsuk's Conjecture) $\mathrm{Bor}(\mathbb{R}^n) \geqslant (1+\varepsilon)^{\sqrt{n}}$.

Proof. The proof can be found in Rob's book from 2010 with Imbuzeiro, but follows the same paradigm. □

2 Ramsey Theory

Just for sake of completeness, we define

$$R(k, l) = \min\{n : K_k \not\subset G \subset K_n \Rightarrow \alpha(G) \geqslant l\}.$$

In coloring terms, it's the smallest n such that any red and blue coloring of $E(K_n)$ contains a red K_k or a blue K_l.

2.1 Bounds on R(3,k)

We have two very simple bounds that are easy to prove,

$$\left(\frac{k}{\log(k)}\right)^{3/2} \leqslant R(3, k) \leqslant \binom{k+1}{2}.$$

The upperbound follows easily from the Erdös-Szekeres proof.

Theorem 2.1. $R(3,k) \leqslant \binom{k+1}{2}$

Proof. Let G on n-vtx be a triangle-free with no independet set of size k, we know every neighborhood is an independent set so $\Delta(G) < k$. Now we greedily get one vtx at a time removing their neighbors. Pick v_1 and $N(v_1)$ with $|N(v_1)| \leqslant k-1$, then pick v_2, notice that the neighborhood of v_2 (after removing v_1 and its neighbors) has size at most $k-2$, if not $\{v_1\} \cup N(v_2)$ would be a independent set of size k. We proceed like this for $v_1, v_2, \dots v_t$, where each step we discard at most $k-(i-1)$ vtxs. Because we can't do this process k times, for some $i < k$ we must fail (meaning there are no vertex left), so $n = |V(G)| \leqslant \sum_{j=0}^{k-2} k-j \leqslant \binom{k+1}{2} - 1$. □

The lowerbound is a beautiful application of the probabilistic method. The main idea is to choose $G(n,p)$ as to have few triangles (linearly small in n) and relatively low independence number.

Theorem 2.2.

$$R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^{3/2}$$

We need a **very important** lemma first.

Lemma 2.3. With high probability, the independence number of $G(n,p) \leqslant 2\log(n)/p$. That is

$$\mathbb{P}(\alpha(G(n,p)) \geqslant 2\log(n)/p + 1) = 1 - o_{(n,p(n))}(1)$$

where p can be taken polynomially small in n.

Proof.

$$\begin{aligned}
\mathbb{P}(\alpha(G(n,p)) \geqslant k) &\leqslant \mathbb{E}[\#\text{independent k sets} \in G(n,p)] \\
&= \binom{n}{k}(1-p)^{\binom{k}{2}} \leqslant \left(\frac{en}{k}\right)^k \cdot e^{-pk(k-1)/2} \\
&= \left(\frac{en}{e^{p(k-1)/2}k}\right)^k
\end{aligned}$$

Taking $k \geqslant 2\log(n)/p + 1$ we get that

$$\mathbb{P}(\alpha(G(n,p)) \geqslant k) \leqslant \left(\frac{ep}{2\log(n)}\right)^{2\log(n)/p} \to 0$$

very fast even! (faster than any polynomial). □

The strategy will be to remove triangles from $G(n,p)$ by removing one vertex from each. This cannot increase the independence number, so we hope the resulting graph still has many vertices to yield a Ramsey construction.

So let's see what this alteration may give us.

Proof. (Of the Theorem) We choose p small such that the number of triangles is no larger than a small linear in n (say $n/2$).

$$\mathbb{E}[\#K_3 \subset G(n,p)] = \binom{n}{3}p^3 = (1+o(1))\frac{(pn)^3}{6}$$

We want

$$\mathbb{P}(\#K_3 \in G(n,p) \geqslant n/2) < (1+o(1))\frac{p^3n^2}{3} < 1/2$$

so taking $p = \varepsilon n^{-2/3}$ for some small ε we get the inequality. But what does this p gives in independence number? We have w.h.p (say bigger than one half)

$$\alpha(G(n,p)) \leqslant 2\varepsilon^{-1}\log(n)n^{2/3}.$$

So $\mathbb{P}(K_3(G(n,p)) < n/2) > 1/2$ and $\mathbb{P}(\alpha(G(n,p) < C\log(n)n^{2/3})) > 1/2$, meaning there exists G satisfying both statements. Remove one vtx from each triangle in G, we are left with a triangle-free graph G' with at least $n/2$ vtxs and $\alpha(G) < C\log(n)n^{2/3}$. Setting $k = C\log(n)n^{2/3}$, we get that

$$V(G') = n/2 \geqslant c\left(\frac{k}{\log(k)}\right)^{3/2}$$

□

Now for a better (and astounishingly beautiful) bound.

Theorem 2.4.

$$R(3,k) \geqslant \left(\frac{k}{\log(k)}\right)^2.$$

The idea of this bound is removing edges. Maybe we can kill the triangles by removing few edges and not increase the independence number too much. For this we will need a beautiful lemma.

Lemma 2.5. (Erdös-Tetali) Let $\{A_1, A_2, \dots A_m\}$ be a family of events in some probability space (Ω, P). Let $\mathcal{E}_t$ be the event that at least t different independent A_i happen. We have that

$$\mathbb{P}(\mathcal{E}_t) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

Proof. We get this one from just opening the terms

$$\mathbb{P}(\mathcal{E}_t) = \mathbb{P}\left(\bigcup_{\substack{i_1<\dots<i_t \\ A_{i_j}\not\sim A_{i_k}}} \bigcap_{j=1}^{t} A_{i_j}\right) \leqslant \sum_{\substack{i_1<\dots<i_t \\ A_{i_j}\not\sim A_{i_k}}} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \sum_{i_1<\dots<i_t} \prod_{j=1}^{t} \mathbb{P}(A_{i_j}) \leqslant \frac{1}{t!}\left(\sum_{i=1}^{m} A_i\right)^t.$$

□

The second ingredient is noting that if you're a bit bigger (say a constant times) than the independent size, you already have many edges (at least half of what is expected).

Lemma 2.6. Let $k \geqslant C\log(n)/p$, then

$$\mathbb{P}\left(\exists \text{ k-set } S, e(G(n,p)[S]) < p\binom{k}{2}/2\right) = o(1)$$

for some $C > 0$.

Proof. Let $\mathcal{E}$ be the event that there exists a k-set S with $e(G(n,p)[S]) < p\binom{k}{2}/2$. We will do a union bound and use Hoeffding (or Chernoff) to bound its probability.

Let S be any k-set. Note that (letting $B(n,p)$ be the binomial with parameters n and p) we have

$$e(G(n,p)[S]) \equiv B\left(\binom{k}{2}, p\right).$$

So by Hoeffding and its bounds on the binomial [A.5] (need to add proofs for these inequalities),

$$\mathbb{P}\left(e(G(n,p)) < \frac{p}{2}\binom{k}{2}\right) = \mathbb{P}\left(\left|B\left(\binom{k}{2}, p\right) - p\binom{k}{2}\right| \geqslant \frac{p}{2}\binom{k}{2}\right) \leqslant e^{-cp\binom{k}{2}}$$

for some $c > 0$.

Now, doing the union bound over k-sets we find that (taking $C > 3c^{-1}$)

$$\mathbb{P}(\mathcal{E}) \leqslant \binom{n}{k} e^{-cp\binom{k}{2}} \leqslant \left(\frac{en}{e^{cp(k-1)/2}k}\right)^k \leqslant \left(\frac{e}{k}\right)^k \to 0.$$

□

The idea to prove theorem 2.4 is to remove a maximal collection of edge-disjoint triangles from $G(n,p)$, (notice that this makes it triangle free). We use lemma 2.5 to show that this procedure won't remove too many edges from any set S with size $C\log(n)/p$.

Proof. (Of Theorem 2.4) Let S be a set of size $k = C\log(n)/p$, for some p we will determine later. Let $\{T_1, \dots T_m\}$ be the set of all triangles that intersect with at least a pair in S. Note that $m \leqslant n\binom{|S|}{2}$. Let $\mathcal{E}_S$ be the event that a edge disjoint set of these triangles with size bigger or equal to $\frac{p}{6}\binom{k}{2}$ are in $G(n,p)$. By [2.5], we can bound $\mathbb{P}(\mathcal{E}_S)$,

$$\mathbb{P}(\mathcal{E}_S) \leqslant \frac{1}{[\frac{p}{6}\binom{k}{2}]!}\left(\sum_{i=1}^{m} \mathbb{P}(T_i \in G(n,p))\right)^t \leqslant \left(\frac{24e}{pk^2}\right)^{pk^2/24} (p^3nk^2)^{pk^2/12}$$

Replacing k in the equation we get (and putting $C > 24e$)

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{24ep}{C\log^2(n)}\right)^{\log(n)^2/(24p)} (pn\log(n)^2)^{\log(n)^2/(12p)} \leqslant (p^3n^2\log(n)^2)^{\log(n)^2/(24p)}$$

So, if we take $p = \varepsilon n^{-1/2}$ (might try to optimize more later), we already find that

$$\mathbb{P}(\mathcal{E}_S) \leqslant \left(\frac{\log(n)}{n^{1/2}}\right)^{n^{1/2}\log^2(n)} \ll \left(\frac{C\log(n)}{n^{1/2}}\right)^{C\log(n)n^{1/2}} \leqslant \binom{n}{k}^{-1}$$

So $\mathcal{E}_S$ does not happen for any S with high probability. As w.h.p S of size k have more than $\frac{p}{2}\binom{k}{2}$ edges, and no maximal disjoint triangle family intersect too much with S. We can remove a maximal edge disjoint triangle family and not make any set of size k independent. This yields a triangle free graph on n vtxs with independence number less than $C\log(n)n^{1/2}$, so

$$R(3,k) \geqslant c\left(\frac{k}{\log k}\right)^2.$$

□

Now let's find an upperbound of the right order.

Theorem 2.7. (Ajtai, Komlós, Szemerédi)

$$R(3,k) \leqslant C\frac{k^2}{\log k}$$

The theorem will be immediately implied by the following lemma.

Lemma 2.8. Let G be K_3-free with $\Delta(G) \leqslant d$, then

$$\alpha(G) \geqslant \frac{cn}{d}\log(d).$$

for some universal constant $c > 0$.

Proof. (Of the theorem assuming Lemma 2.8) Notice that G is K_3 free and its independence number is less than k, then, as every neighborhood is a independent set, we have $\Delta(G) < k$. Applying the lemma we get that

$$k > \alpha(G) \geqslant \frac{cn}{k}\log(k)$$

which implies

$$n \leqslant C\frac{k^2}{\log(k)}.$$

□

The proof of the lemma is exceptionally fun. The main idea is to take a uniform random independent set in G, and show that in expectation, its size is bigger than $\frac{cn}{d}\log(d)$ - which is very unnintuitive.

Proof. (Of Lemma 2.8) Let $I \in \mathcal{I}(G)$ be a uniformly chosen random independent set of G. For each $v \in V(G)$, define the random variable

$$X_v = d \cdot \mathbb{1}[v \in I] + \sum_{u \in N(v)} \mathbb{1}[u \in I].$$

Note that

$$\mathbb{E}\sum_{v \in V(G)} X_v = d \cdot \mathbb{E}|I| + \sum_{v \in V(G)}\sum_{u \in N(v)} \mathbb{P}(u \in I) = d \cdot \mathbb{E}|I| + \sum_{u \in V(G)} d(u)\mathbb{P}(u \in I) \leqslant 2d \cdot \mathbb{E}|I|.$$

So we have that $\mathbb{E}|I| \geqslant \frac{1}{2d}\mathbb{E}[\sum_{v\in V(G)} X_v]$. To prove the lemma, it would suffice to show then that $\mathbb{E}X_v \geqslant \log(d)$ for all v (as summing it would yield the result).

So let's lowerbound $\mathbb{E}X_v$. The idea is to condition it on something and find a bound that works **FOR EVERY** condition we take (which seems insane).

Fix v, let's divide the world into $U = \{v\} \cup N(v)$ and $W = V(G) \setminus U$ (both depend on v). Notice that the value of X_v only changes for elements in U that appear in I. So fix an independent set S of W, and let's calculate $\mathbb{E}[X_v | I \cap W = S]$.

From the fact that I has to be an independent set, we know it cannot have any neighbors of elements in S. So let $Y = N(v) \cap N(S)$ and let $X = N(v) \setminus Y$ (note $v \notin N(S)$). As I is uniformly chosen, we now know that $I \cap U$ (conditioned on $I \cap W = S$), is a uniformly chosen independent set of $\{v\} \cup X$. Let $|X| = t$, remembering that X is an independent set, we find

$$\begin{aligned}\mathbb{E}[X_v | I \cap W = S] &= \mathbb{E}[X_v | v \in I, I \cap W = S] \cdot \mathbb{P}(v \in I | I \cap W = S) \\ &\quad + \mathbb{E}[X_v | v \notin I, I \cap W = S] \cdot \mathbb{P}(v \notin I | I \cap W = S) \\ &= \frac{1}{2^t+1} d + \frac{2^t}{2^t+1}\frac{t}{2}\end{aligned}$$

where we used the fact that $I \cap U$ is an uniform independent set of the star $G[\{v\} \cup X]$. Now (here comes the crazy part), notice that independent of what $t = |X|$ is, we have

$$\frac{1}{2^t+1} d + \frac{2^t}{2^t+1}\frac{t}{2} \geqslant \frac{\log(d)}{8}.$$

just take $t < \log(d)/2$ and $t \geqslant \log(d)/2$. But this means

$$\mathbb{E}X_v = \mathbb{E}[\mathbb{E}[X_v | I \cap W(v)]] \geqslant \frac{\log(d)}{8}$$

yielding the result. □

2.2 Bounds on R(4,k)

By doing vertex deletion and Erdös-Szekeres, we may easily find (we need $p^6 n^4 < n$, so $p = \varepsilon n^{-1/2}$)

$$c\left(\frac{k}{\log(k)}\right)^2 \leqslant R(4,k) \leqslant k^3.$$

Doing the Erdös-Tetali method [2.5] (we need $p^6 n^4 < pn^2$, so $p = \varepsilon n^{-2/5}$) we get also

$$c\left(\frac{k}{\log k}\right)^{5/2} \leqslant R(4,k).$$

But we can do better on the upperbound by doing a similar strategy as in the $R(3,k)$ case. The next lemma Rob stated, but I don't think we will use it.

Lemma 2.9. Let G on n vtx be K_4-free and $\Delta(G) \leqslant d$, then

$$\alpha(G) \geqslant \frac{cn}{d}\frac{\log(d)}{\log\log(d)}.$$

Proof. The proof of this is similar to lemma 2.8, we must however pick a uniformly independent set from the now K_3-free X. (I will try to do the computations more carefully later) $\square$

To get an even better bound, we will need some kind of stability setting in these super sparse graphs.

Lemma 2.10. (AKSz) Let G be a graph on n-vtx with $\Delta(G) \leqslant d$ and $k_3(G) \leqslant (nd^2)/\lambda$, then

$$\alpha(G) \geqslant c\frac{n}{d}\log(\lambda)$$

where $c > 0$ is an absolute constant.

With it we can prove the following strong bound,

Theorem 2.11.

$$R(4,k) \leqslant C\frac{k^3}{\log(k)^2}.$$

Proof. (Assuming [2.10]) Let G be n-vtx, K_4-free graph with $\alpha(G) < k$, we know by Erdös-Szekeres that $\Delta(G) < R(3,k)$. How many triangles can G have? Pick a vertex v and let $S = N(v)$. We know that for any vertex $u \in S$, $N_{G[S]}(u)$ is independent (because G is K_4-free) and so $d_{G[S]}(u) < k$. This implies that

$$k_3(G) = \frac{1}{3}\sum_{v\in V(G)} e(G[N(v)]) \leqslant \frac{nk^2}{6}$$

by our bound on $\Delta(G[S])$.

So applying [2.10] with $d = R(3,k)$ and $\lambda = (k/\log(k))^2$ yields

$$k > \alpha(G) > c\frac{n\log(k)^2}{k^2}$$

which yields $n < Ck^3/\log(k)^2$. $\square$

So now it suffices to prove 2.10.

Proof. (Of the AKSz lemma) The idea will be to do some vertex deletion and apply 2.8, but we don't really know how so we do it randomly.

To easy up the calculation (it only changes the constant in front), suppose $\Delta(G) \leqslant d$ and $k_3(G) \leqslant nd^2/\lambda^3$. Take a q-random subset Q of $V(G)$, we can estimate some statistics, we find:

(i) $\mathbb{E}|Q| = qn$

(ii) $\mathbb{E}e(G[Q]) = q^2e(G) \leqslant q^2nd/2$

(iii) $\mathbb{E}k_3(G[Q]) = q^3k_3(G) \leqslant (q/\lambda)^3nd^2$

By taking $q = \lambda/d$ and using Markov to remove triangles and high degree vtxs, we find a set of size at least $qn/6$ with no triangles and $\Delta(G[Q']) < 3q\Delta(G)$. Applying 2.8 to this set yields the result.

(Formally) We have very good concentration on $|Q|$ (as long as $qn \to \infty$), so applying Chernoof and Markov (twice) we see that

$$\begin{aligned}\mathbb{P}\left(\left(|Q| > \frac{qn}{2}\right) \wedge \left(e(G[Q]) < 3q^2 e(G)\right) \wedge \left(k_3(G[Q]) < \frac{3q^3nd^2}{\lambda^3}\right)\right) \\ = 1 - \mathbb{P}\left(\left(|Q| < \frac{qn}{2}\right) \vee \left(e(G[Q]) > 3q^2 e(G)\right) \vee \left(k_3(G[Q]) > \frac{3q^3nd^2}{\lambda^3}\right)\right) \\ \geqslant 1 - o(1) - 1/3 - 1/3 > 0\end{aligned}$$

So we find a $G' = G[Q]$ for some good Q satisfying all these properties. G' satisfies:

(i) $v(G') \geqslant \frac{qn}{2}$

(ii) $e(G') \leqslant 3q^2 e(G) \leqslant 3q^2 nd/2$

(iii) $k_3(G') \leqslant \frac{3q^3nd^2}{\lambda^3}$

By removing at most half of the vertices of G' (using Markov) and killing a vertex from each triangle, we find triangle free G'' with $v(G') \geqslant \frac{qn}{4} - k_3(G') \geqslant \frac{qn}{8}$ and $\Delta(G'') \leqslant 6q^2nd/2$. □

This seems like a very interesting strategy for off-diagonal Ramsey. We did the case $l = 3$, $l = 4$, so a very natural conjecture would be:

Theorem 2.12.

$$R(l,k) \leqslant C\frac{k^{l-1}}{(\log k)^{l-2}}$$

Proof. We prove it by induction. The case $l = 3, 4$ were already done. Let's follow the proof of [2.11]. Suppose G is an n-vtx, K_l-free graph with $\alpha(G) < k$. We must have $\Delta(G) < R(l-1,k) \leqslant d$. How many triangles can it have? We know that if $u \in N(v)$, then $|N(u) \cap N(v)| < R(l-2,k)$. So, doing the same bound as before, we see that $k_3(G) \leqslant ndR(l-2,k)/3$. So, to use 2.10 we must have

$$\frac{ndR(l-2,k)}{3} \leqslant \frac{nd^2}{\lambda}$$

so

$$\lambda \leqslant \frac{3d}{R(l-2,k)}$$

and (by induction), it suffices that λ satisfies

$$\lambda \leqslant \frac{cd(\log k)^{l-2}}{k^{l-1}}$$

applying 2.10 with $d = \frac{k^{l-1}}{(\log k)^{l-2}} \geqslant R(l-1,k) > \Delta(G)$, we have by hypohesis,

$$\alpha(G) \geqslant \frac{n}{d}\log(\lambda) \geqslant C_l\frac{n\log(k)^{l-2}}{k^{l-2}}$$

and therefore,

$$n \leqslant C_l \frac{k^{l-1}}{(\log k)^{l-2}}$$

□

This proof can also be found in Rob's ICM manuscript. [https://arxiv.org/pdf/2601.05221].

2.3 Bounds on R(3,3,k)

So what are some easy bounds we can get on this? An easy lowerbound is $R(3,k)$, by doing Erdös-Szekeres, we can get the multinomial bound as well

$$R(3,k) \leqslant R(3,3,k) \leqslant 2kR(3,k) \leqslant C\frac{k^3}{\log k}.$$

We can do a bit better on the upperbound by counting triangles and applying [2.10]!

Theorem 2.13.

$$R(3,3,k) \leqslant \frac{k^3}{(\log k)^2}$$

Proof. Suppose there is a coloring χ on K_n that shows that $R(3,3,k) \geqslant n$. Let $G = E_R(\chi(K_n)) \cup E_B(\chi(K_n))$ that is, G is the graph of red and blue edges. Note that $\alpha(G) < k$, as the coloring had no green K_k. To apply [2.10], we must calculate $k_3(G)$ and we know it does not have any full red, or any full blue triangles. So let's bound the number of $\{R,R,B\}$ triangles in G (the number of $\{B,B,R\}$ by simmetry will be the same). One way to count $\{R,R,B\}$ triangles is by picking the cherry vertex.

$$k_{\{R,R,B\}}(G) = \sum_{v \in G} e_B(G[N_R(v)])$$

By Erdös-Szekeres, we know that $N_R(v)$ is red-edge-free and $|N_R(v)| \leqslant R(3,k)$ for any vertex v. We also have that for any $u \in N_R(v)$, $N_B(u) \cap N_R(v)$ is a independent set, by the previous observation and the fact that we are blue triangle free. So $d_B[G[N_R(v)]](u) < k$ (here this monstrosity means that the blue degree of u in $N_R(v)$ is less than k). This implies that $e_B(G[N_R(v)]) < R(3,k)k/2$ for any v and therefore that

$$k_{\{R,R,B\}}(G) < \frac{nR(3,k)k}{2}$$

so the total number of triangles is also of order $nk^3/\log k$. Applying [2.10] with $d = \frac{Ck^2}{\log k} \geqslant R(3,k)$ and this $k_3(G) < ndk$ we get that we can use $\lambda = Cd/k \geqslant \frac{Ck}{\log k}$. So we get

$$k > \alpha(G) \geqslant \frac{n}{d}\log k \geqslant \frac{C'n(\log k)^2}{k^2}$$

which implies $n \leqslant Ck^3/(\log k)^2$. □

Next class we will see a very cool lowerbound by Alon, who observed the following lemma:

Lemma 2.14. If there exists K_3-free G on n vtx with number of independent k sets of G less than $\binom{n}{k}^{1/2}$ then $R(3,3,k) > n$.

Proof. Embed G randomly in K_n with color red, do it again in color blue. By the union bound, no independent k set survive (so we win by painting non edges with green). □

We will turn for a pseudo-random construction to prove the existence of a graph that satisfies [2.14]. For this, we need a good definiton of what pseudo-random means.

Definition 2.15. We say that a n-vtx graph G is (p,β)-jumbled if $e(G) = p\binom{n}{2}$ and for every $S, T \subset V(G)$, we have that

$$|e(S,T) - p|S||T|| \leqslant \beta\sqrt{|S||T|}.$$

where here $e(S,T)$ is the number of edges that start in S and end in T. Edges in $S \cap T$ are counted twice.

The most pseudo-random graph (the best β you can get) is with $G(n,p)$, I don't know anyway to prove this now, but let's calculate how jumbled $G(n,p)$ is.

Theorem 2.16. $G(n,p)$ is (p',β)-jumbled with $p' = e(G(n,p))/\binom{n}{2} = (1+o(1))p$ and $\beta = O(\sqrt{pn})$ w.h.p.

Proof. We use Chernoff inequality [A.5], to note that for absolute $C, c > 0$ and taking A suffciently big,

$$\mathbb{P}(|e(S,T) - p|S||T|| \geqslant A\sqrt{pn|S||T|}) \leqslant C\exp(-cA^2n) \ll 4^{-n},$$

so $G(n,p)$ is $\beta = A\sqrt{np}$-jumbled w.h.p by the union bound. □

This yields the following definition

Definition 2.17. A graph is said to be **optimally pseudorandom** if it is (p,β)-jumbled with $\beta = O(\sqrt{pn})$.

The following question is at the core of this method. How many independent k-sets can a (p,β)-jumbled graph have? We now have a deterministic fact about our graph and we can try to count. I won't do it here, but the idea is to count sequences $(v_1,\ldots,v_k)$ of vertices that can make a independent set.

Using the (p,β) jumbled condition, we can count how many "bad" vertices there are when trying to grow the sequence. We can make only a few "good" steps (extension steps where our non-neighborhood decreases with correct size), because they reduce the non-neighborhood by a $(1-p)/2$ ratio which is just too fast. And each "bad" step (steps which don't reduce the non-neighborhood much) we are obliged to choose from a small set. So counting everything yields a result.

The following lemma will be stated without proof, it is an algebraic construction.

Lemma 2.18. (Quadrangle) There exists a n-vertex, $n^{1/3}$-regular, $n^{1/3}$-uniform hypergraph H that satisfies:

(i) $\forall e_1, e_2 \in E(H), \quad |e_1 \cap e_2| \neq 2.$

(ii) $\forall e_1, e_2, e_3 \in E(H)$ it does not hold that $|e_1 \cap e_2| = |e_2 \cap e_3| = |e_1 \cap e_3| = 1$.

Putting a randomly embedded $K_{t,t}$ with $t = n^{1/3}/2$ on each edge we get a K_3-free, optimally pseudorandom graph where p is the maximum it can be, $p = n^{-1/3}$. (As Rob pointed out an optimally pseudorandom, triangle-free graph must have density at most $p = n^{-1/3}$).

A (Sub)Exponential Tailbounds

I just took these from my notes in random matrices.

Lemma A.1. (Hoeffding's Lemma) Let X be a scalar r.v taking values in $[a, b]$. Then for any $t > 0$

$$\mathbb{E}e^{tX} \leqslant e^{t\mathbb{E}X}(1 + O(t^2\sigma^2 \exp(O(t(b-a))))). \tag{1}$$

In particular

$$\mathbb{E}e^{tX} \leqslant e^{t\mathbb{E}X} \exp(O(t^2(b-a)^2)). \tag{2}$$

Proof. Normalize X as to $\mathbb{E}(X) = 0$ and $b - a = 1$. And consider the Taylor expansion of e^{tX},

$$e^{tX} = 1 + tX + O(t^2X^2 \exp(O(t)))$$

taking expectation yields the first result. The second one follows from assymptotics of the exponential function. □

Theorem A.2. (McDiamird's Theorem) Let $(R_i)_{i=1}^n$ be a tuple of ranges and $X = (X_i)_{i=1}^n$ a random vector taking values in their product. Let $F : \prod_{i=1}^n R_i \to \mathbb{R}$ be a coordinate-wise bounded function (this is a strong Lipchitz condition). That is changing any coordinate x_i may change at most c_i the total value. Then

$$\mathbb{P}(|F(X) - \mathbb{E}F(X)| > \lambda\sigma) \leqslant C \exp(-c\lambda^2)$$

for some $C, c > 0$ and $\sigma^2 = \sum_{i=1}^n c_i^2$.

Proof. By simmetry, we may only consider $\mathbb{P}(F(X) - \mathbb{E}F(X) > \lambda\sigma)$. We turn to Hoeffding's Lemma [A.1], consider

$$\mathbb{E}\exp(tF(X)) = \mathbb{E}\big[\mathbb{E}[\exp(tF(X))|X_1, \dots X_{n-1}]\big],$$

we should try to understand $\mathbb{E}[\exp(tF(X))|X_1, \dots X_{n-1}]$. We have that

$$\mathbb{E}[\exp(tF(X))|X_1, \dots X_{n-1}] = \mathbb{E}[\exp(t(F(X) - \mathbb{E}(F))) \exp(t\mathbb{E}F(X))|X_1, \dots X_{n-1}]$$

applying (2) to the first product, (here we are conditioning on $X_1, \dots X_n$)

$$\mathbb{E}[\exp(t(F(X) - \mathbb{E}(F))) \exp(t\mathbb{E}F(X))|-] \leqslant \exp(O(t^2c_n^2))\mathbb{E}[\exp(t\mathbb{E}F(X))|-]$$

and we can iteratively apply the theorem to the new coordinate-wise bounded function

$$G(X_1, \dots X_{n-1}) = \mathbb{E}[F(X)|X_1, \dots X_{n-1}].$$

Notice G has the same coefficients as F, because

$$\begin{aligned}|G(y, x_2 \ldots, x_{n-1}) - G(z, x_2 \ldots, x_{n-1})| &= \left| \int_{R_n} F(y, \ldots, x_n) dx_n - \int_{R_n} F(z, \ldots, x_n) dx_n \right| \\ &= \left| \int_{R_n} F(y, \ldots, x_n) - F(z, \ldots, x_n) dx_n \right| < c_1,\end{aligned}$$

and here we can take any other coordinate instead of the first. □

Corollary A.3. (Hoeffding's inequality) Let $(X_i)_{i=1}^n$ be real r.v's taking value in bounded intervals $[a_i, b_i]$. Then, if $S = \sum_{i \leqslant n} X_i$, we have

$$\mathbb{P}(|S - \mathbb{E}S| > \lambda\sigma) < C \exp(-c\lambda^2)$$

for some $C, c > 0$, where $\sigma^2 = \sum_{i \leqslant n} (b_i - a_i)^2$.

Proof. Apply [A.2] with $F(X_1, \ldots X_n) = X_1 + \cdots + X_n$. □

Theorem A.4. (Chernoff) Let $(X_i)_{i=1}^n$ be independent scalar random variables with $|X_i| < K$ a.s, with mean μ_i, and variance σ_i^2. Then for any $\lambda > 0$, one has

$$\mathbb{P}(|S_n - \mu| \geqslant \lambda\sigma) \leqslant C \exp(-c \min(\lambda^2, \frac{\lambda\sigma}{K}))$$

for some absolute $C, c > 0$, where $S_n = \sum_{i=1}^n X_i$, $\mu = \sum_{i=1}^n \mu_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$.

Proof. The proof is again the use of the exponential moment with lemma [A.1]. Suppose we normalize each variable so that $K \leqslant 1$ and $\mu_i = 0$, then $\mu = 0$ and for any $0 \leqslant t \leqslant 1$,

$$\mathbb{E}(\exp(tS_n)) = \prod_{i=1}^n \mathbb{E}(\exp(tX_i)) = \prod_{i=1}^n (1 + O(t^2 \sigma_i^2 \exp(O(t)))).$$

using the usual $1 + x \leqslant e^x$ ineq, we find

$$\mathbb{E}(\exp(tS_n)) \leqslant \exp(t\mathbb{E}(S_n)) \exp(O(t^2) \exp(O(t))\sigma^2) = \exp(O(t^2\sigma^2)),$$

now we use Markov and optimize on t following the restraint on $0 \leqslant t \leqslant 1$. (Suppose the hidden constant is D) we have that

$$\mathbb{P}(S_n \geqslant \lambda\sigma) \leqslant \exp(Dt^2\sigma^2 - t\lambda\sigma)$$

which yields the result after optimizing t following the constraints. The other inequality is equivalent.

□

Corollary A.5. (Tailbounds for the Binomial distribution) Let $B(n, p)$ be the binomial distribution. We have that

$$\mathbb{P}(|B(n, p) - np| \geqslant \delta np) \leqslant C \exp(-c\delta^2 (np)^2)$$

for some absolute C, c.

Proof. Note that

$$\mathbb{P}(|B(n,p) - np| \geqslant \lambda\sqrt{np}) \leqslant \mathbb{P}(|B(n,p) - np| \geqslant \lambda\sqrt{np(1-p)}) \leqslant C\exp(-c\lambda^2)$$

by A.4. Applying it with $\lambda = \delta\sqrt{np}$ yields the result. □

Theorem A.6. (Azuma's Inequality) Let $X_1, X_2, \ldots X_n$ be a sequence scalar r.v's with $|X_i| \leqslant 1$ almost surely. Assume also that we have the martingale difference property

$$\mathbb{E}(X_i | X_1, \ldots, X_{i-1}) = 0$$

almost surely for all $i = 1, \ldots, n$. Then for any $\lambda > 0$, the sum $S_n := X_1 + \cdots + X_n$ satisfies

$$\mathbb{P}(|S_n| \geqslant \lambda\sqrt{n}) \leqslant C\exp(-c\lambda^2)$$

for some absolute $C, c > 0$.

Proof. We prove this by calculating exponential moments and using Hoeffding's Lemma again. Note that $S_n = S_{n-1} + X_n$ so that

$$\mathbb{E}(e^{tS_n}) = \mathbb{E}(e^{tS_{n-1}} e^{tX_n}).$$

By taking the expectation of conditioning on the first $n-1$ values, we get that

$$\mathbb{E}(e^{tS_n}) = \mathbb{E}\mathbb{E}(e^{tS_{n-1}} e^{X_n} | X_1, \ldots, X_{n-1}) = \mathbb{E}[e^{tS_{n-1}} \mathbb{E}(e^{tX_n} | X_1, \ldots, X_{n-1})]$$

Now, by A.1, we have that $\mathbb{E}(e^{tX_n} | X_1, \ldots, X_{n-1}) \leqslant e^{O(t^2)}$, so by iterating we get the result.

□